Comparison of Generative Models on the ArtBench-10 Dataset

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Abstract. This paper presents the implementation and comparative evaluation of three generative image model families — a Variational Autoencoder (VAE), a Deep Convolutional Generative Adversarial Network (DCGAN), and a Denoising Diffusion Probabilistic Model (DDPM) — trained on the ArtBench-10 dataset (32×32 pixels, 10 artistic styles). Following a two-phase protocol, all models were first developed and evaluated on a 20% data subset, then retrained on the full 50,000-image dataset. Quantitative evaluation uses Fréchet Inception Distance (FID) and Kernel Inception Distance (KID), reported as mean $\pm$ std over 10 seeds (5,000 samples each). On the full dataset, the DCGAN achieved the best results (FID $= 30.46 \pm 0.57$; KID $= 0.0032 \pm 0.0001$), followed by the DDPM (FID $= 37.91 \pm 0.37$; KID $= 0.0062 \pm 0.0002$) and the VAE (FID $= 140.37 \pm 0.43$; KID $= 0.0314 \pm 0.0004$).

Keywords: Generative Models · VAE · DCGAN · DDPM · ArtBench-10 · FID · KID

1 Introduction

Generative modelling of images has become a central research area, with applications ranging from data augmentation to creative content synthesis. Three paradigms have emerged as dominant: Variational Autoencoders (VAE) [6], Generative Adversarial Networks [8], and Diffusion Models [3], each offering distinct trade-offs between sample fidelity, diversity, training stability, and inference cost.

The ArtBench-10 dataset [7] provides a curated benchmark for generative models in the visual arts domain. Artworks are particularly challenging: complex textures, high style variability, and the need for both visual realism and stylistic coherence make this a demanding generative task.

This work implements and compares all three model families under a two-phase protocol: Phase 1 trains and evaluates on a 20% subset for rapid iteration; Phase 2 retrains on the full dataset for final comparison. All three models were retrained in Phase 2, exceeding the minimum requirement.

The paper is organised as follows: Section 2 describes the problem and dataset; Section 3 details the three model architectures; Section 4 describes the experimental setup; Section 5 covers Phase 1 results (20% subset); Section 6 covers Phase 2 results (full dataset); and Section 7 concludes.

2 Problem Description

The goal is unconditional image generation: given a dataset of artwork images, a generative model must learn the data distribution and produce new samples that are visually plausible and statistically consistent with training data.

Dataset: ArtBench-10

ArtBench-10 [7] is a curated benchmark for generative modelling in the visual arts domain. Its main characteristics are:

- **Classes:** 10 artistic styles — *Art Nouveau*, *Baroque*, *Expressionism*, *Impressionism*, *Post-Impressionism*, *Realism*, *Renaissance*, *Romanticism*, *Surrealism*, and *Ukiyo-e*.
- **Size:** 50,000 training images and 10,000 test images (5,000/1,000 per class, class-balanced).
- **Resolution:** 32×32 RGB.

Development Protocol

All three models were first trained on a **20% subset** (10,000 images) to allow fast iteration over architectures and hyperparameters. Subsequently all three were retrained on the **full training set** (50,000 images) for the final evaluation phase.

3 Methodology

3.1 Variational Autoencoder (VAE)

The VAE [6] learns a latent representation by maximising the Evidence Lower BOund (ELBO):

$$\mathcal{L}(\theta, \phi;\, x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{\mathrm{KL}}(q_\phi(z|x) \,\|\, \mathcal{N}(0, I))\,, \tag{1}$$

where $q_\phi(z|x) = \mathcal{N}(\mu_\phi, \sigma_\phi^2 I)$ and sampling uses the reparameterisation trick $z = \mu + \sigma \odot \varepsilon$. A β-VAE formulation ($\beta = 0.5$) weights reconstruction more heavily: $\mathcal{L}_\beta = \mathcal{L}_{\mathrm{recon}} - \beta\, D_{\mathrm{KL}}$.

Architecture ($\sim$2.9 M params). *Encoder:* three strided conv. blocks ($3 \to 64 \to 128 \to 256$ channels, stride 2, BatchNorm, LeakyReLU 0.2), then linear heads to $\mu, \log \sigma^2 \in \mathbb{R}^{128}$. *Decoder:* three transposed-conv. blocks ($256 \to 128 \to 64 \to 3$, BatchNorm, ReLU, Sigmoid).

3.2 Deep Convolutional GAN (DCGAN)

GANs [8] train a generator G and discriminator D via:

$$\min_G \max_D \; \mathbb{E}_{x\sim p_{\text{data}}}[\log D(x)] + \mathbb{E}_{z\sim p_z}[\log(1 - D(G(z)))] \,. \tag{2}$$

DCGAN [8] stabilises training via strided convolutions (no pooling), BatchNorm, and LeakyReLU in the discriminator.

Architecture (∼4.1 M params, G+D). The generator maps $z \in \mathbb{R}^{100}$ through four transposed-conv. blocks ($512\times4\times4 \to 256 \to 128 \to 3\times32\times32$, Tanh). The discriminator uses four strided convolutions ($3 \to 64 \to 128 \to 256 \to 1$, Sigmoid, LeakyReLU 0.2). Weights: $\mathcal{N}(0, 0.02)$; label smoothing = 0.9.

3.3 Denoising Diffusion Probabilistic Model (DDPM)

DDPMs [3] define a Markov forward noising process over T steps and train a U-Net ϵ_θ to reverse it:

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\big(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon,\, t\big)\right\|^2\right] . \tag{3}$$

For efficient inference, DDIM [1] sampling uses 100 steps instead of the 1,000 training steps (10× speedup).

Architecture (29.60 M params). A U-Net at three resolution levels ($32\to16\to8$). Each level has two ResBlocks (GroupNorm, SiLU, Dropout 0.1). Self-attention at the 16×16 and 8×8 levels. Timestep encoded via sinusoidal embedding + two-layer MLP, injected into every ResBlock. EMA decay = 0.9995.

4 Experimental Setup

4.1 Data Preprocessing

All images are resized/centre-cropped to 32×32. The VAE normalises to $[0, 1]$; DCGAN and DDPM to $[-1, 1]$. The DDPM applies additional augmentation: random crop (4-pixel margin), horizontal flip ($p = 0.5$), colour jitter (brightness/contrast/saturation ±5%, hue ±2%).

4.2 Training Configuration

Table 1 summarises the main hyperparameters. All experiments used a fixed random seed (42) for reproducibility; training ran on a single NVIDIA GPU with CUDA and mixed-precision (AMP).

4.3 Evaluation Protocol

Both phases share the same protocol: 5,000 generated images vs. 5,000 real test images. **FID** uses InceptionV3 features (2,048-dim, fc → Identity). **KID** is the mean MMD^2 over 50 random subsets of size 100. The full protocol is repeated for 10 seeds (42–51); results are reported as mean ± standard deviation.

Table 1. Training hyperparameters (full-dataset phase; 20% phase uses the same settings).

Hyperparameter	VAE	DCGAN	DDPM
Epochs	100	200	500
Batch size	64	64	128
Optimiser	Adam	Adam	AdamW
Learning rate	10^{-3}	2×10^{-4}	2×10^{-4}
Adam β_1	0.9	0.5	0.9
Latent / noise dim	128	100	—
Noise steps T	—	—	1000
LR schedule	None	None	Warmup+cosine
EMA decay	—	—	0.9995
Total grad. steps	∼78k	∼156k	∼195k
Parameters	∼2.9M	∼4.1M	29.60M
Inference passes	1	1	100 (DDIM)

5 Phase 1 — Development on 20% Subset

All three models were trained and qualitatively tuned on the 10,000-image subset before the full-dataset retraining.

5.1 VAE — 20% Subset

Figure 1 shows the VAE training curves and generated samples on the subset. The total loss decreased from ≈920 to ≈895 over 100 epochs, with the KL divergence stabilising at ≈55, which indicates a well-formed latent space. Generated images display the characteristic blurriness of BCE reconstruction: the decoder averages over plausible sharp images, producing smoothed outputs.

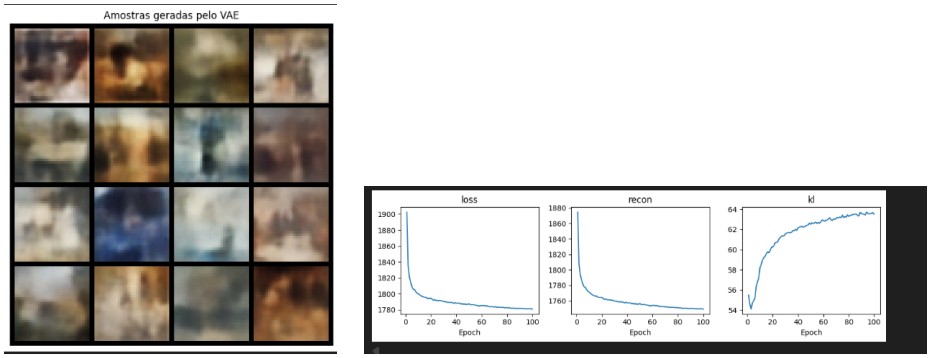

Fig. 1. VAE on 20% subset. Left: 64 generated samples. Right: training curves (total loss, BCE, KL divergence).

5.2 DCGAN — 20% Subset

Figure 2 shows the DCGAN adversarial losses and generated samples. Generator and discriminator losses reached a quasi-stationary equilibrium, reflecting stable adversarial training. Generated images are visibly sharper than VAE outputs; some repetition of motifs is apparent, suggesting mild mode dropping on the limited subset.

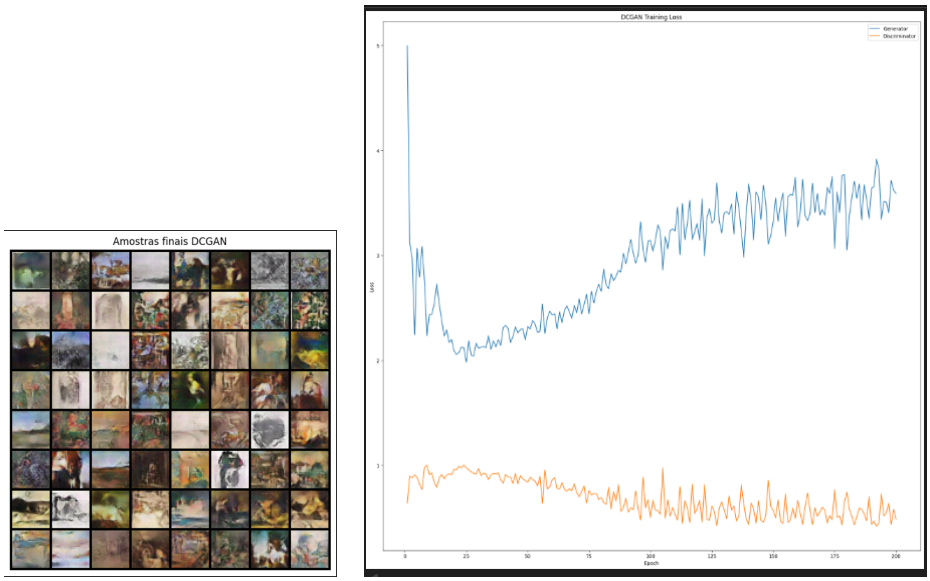

Fig. 2. DCGAN on 20% subset. Left: 64 generated samples. Right: generator ($\mathcal{L}_G$) and discriminator ($\mathcal{L}_D$) loss curves.

5.3 DDPM — 20% Subset

Figure 3 shows the DDPM MSE loss and generated samples. The loss converged smoothly under the warmup+cosine schedule. Despite the limited data, the DDPM produced the most diverse outputs of the three models, covering a wide range of styles and colour palettes — a direct consequence of the iterative stochastic reverse process.

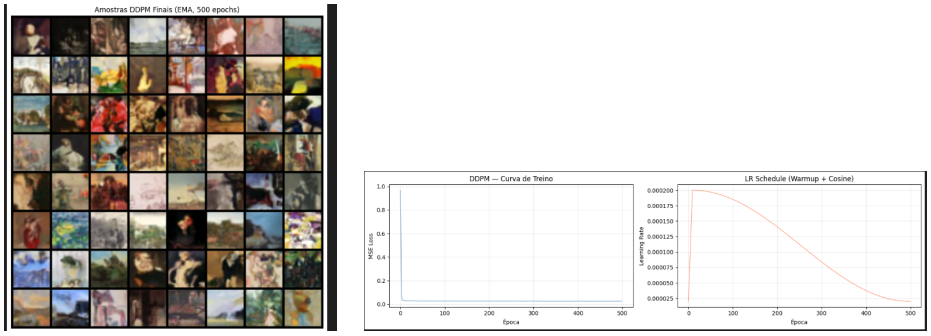

Fig. 3. DDPM on 20% subset. Left: 64 generated samples (DDIM, 100 steps). Right: MSE loss and LR schedule over 500 epochs.

5.4 Quantitative Evaluation — 20% Subset

Table 2 and Figure 4 report FID and KID on the 20% subset across 10 seeds.

Table 2. FID and KID on the 20% subset (mean ± std, 10 seeds). Bold: best value per metric.

Model	**FID** (↓)	**KID** (↓)
VAE	142.72 ± 0.64	0.0305 ± 0.0004
DCGAN	47.38 ± 0.65	$\mathbf{0.0066} \pm \mathbf{0.0001}$
DDPM	$\mathbf{45.91} \pm \mathbf{0.48}$	0.0078 ± 0.0002

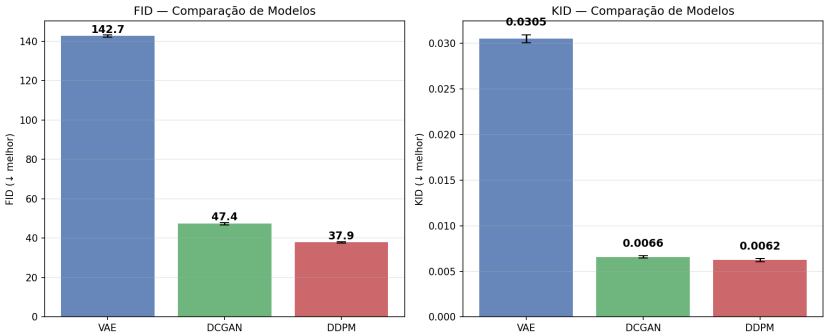

Fig. 4. FID and KID on the 20% subset (10 seeds, error bars = 1 std).

On the subset, DDPM achieved a marginally lower FID (45.91) than DCGAN (47.38), while DCGAN led on KID (0.0066 vs. 0.0078). The VAE lagged by a

wide margin (142.72). The near-parity between DCGAN and DDPM motivated retraining both on the full dataset.

6 Phase 2 — Full Dataset (100%)

All three models were retrained from scratch on the complete 50,000-image training set with identical hyperparameters to Phase 1.

6.1 VAE — Full Dataset

Figure 5 shows VAE training curves and samples on the full dataset. The total loss converged from ≈1840 to ≈1780 over 100 epochs, with the KL term stabilising at ≈65. The 5× data increase did not qualitatively improve image sharpness, confirming that blurriness is a structural limitation of BCE reconstruction rather than a data-quantity issue.

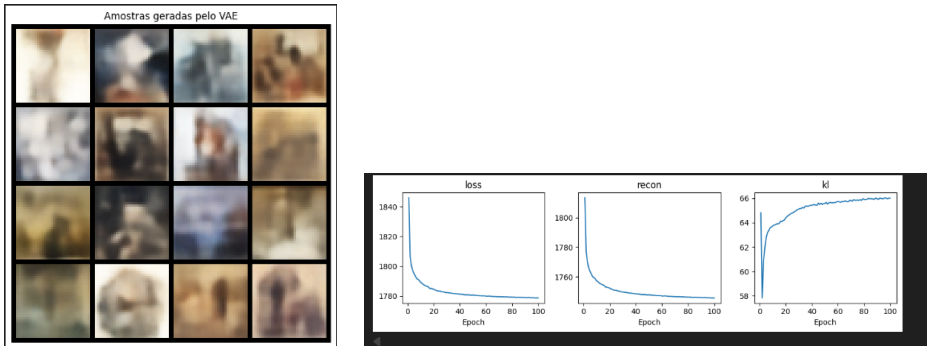

Fig. 5. VAE on full dataset. Left: 64 generated samples. Right: training curves (total loss, BCE, KL divergence).

6.2 DCGAN — Full Dataset

Figure 6 shows DCGAN results on the full dataset. Adversarial losses maintained stable dynamics across 200 epochs. Sample quality improved noticeably compared to Phase 1: images show richer colour palettes and finer texture detail, consistent with the discriminator having access to 5× more real examples against which to sharpen the generator.

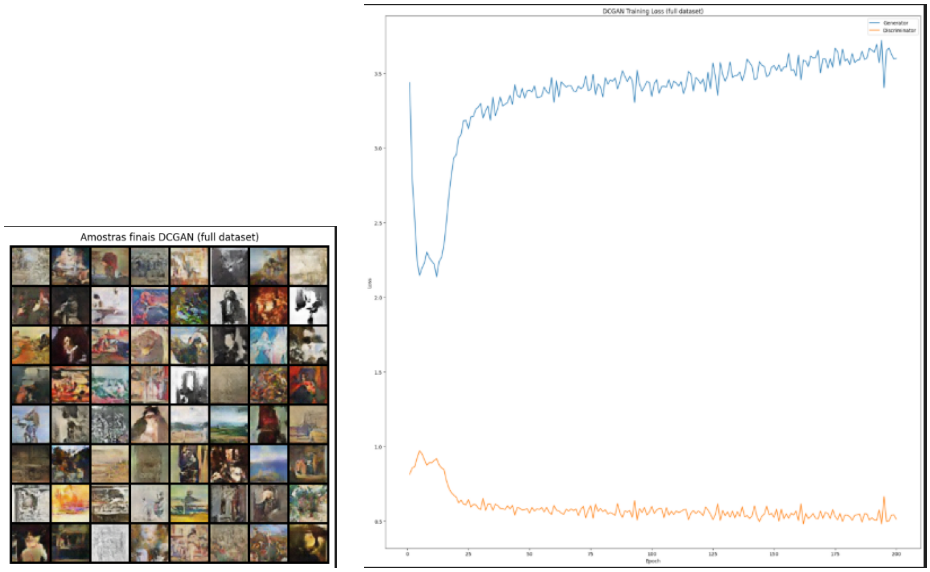

Fig. 6. DCGAN on full dataset. Left: 64 generated samples. Right: generator and discriminator loss curves.

6.3 DDPM — Full Dataset

Figure 7 shows DDPM results on the full dataset. The MSE loss converged smoothly over 500 epochs with the warmup+cosine schedule. Samples retain the high style diversity seen in Phase 1 and show further improvement in colour coherence. However, the FID improvement from Phase 1 to Phase 2 was smaller than for DCGAN, suggesting the adversarial objective is more sensitive to data quantity at 32×32 resolution.

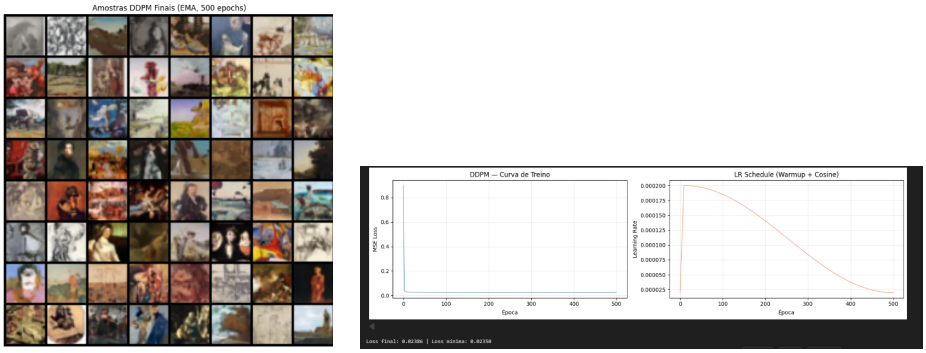

Fig. 7. DDPM on full dataset. Left: 64 generated samples (DDIM, 100 steps). Right: MSE loss and LR schedule over 500 epochs.

6.4 Quantitative Evaluation — Full Dataset

Table 3 summarises the main results; Table 4 provides the full per-seed breakdown; Figure 8 visualises the distributions.

Table 3. FID and KID on the full dataset (mean ± std, 10 seeds). Bold: best value.

Model	**FID** (↓)	**KID** (↓)
VAE	140.37 ± 0.43	0.0314 ± 0.0004
DCGAN	$\mathbf{30.46 \pm 0.57}$	$\mathbf{0.0032 \pm 0.0001}$
DDPM	37.91 ± 0.37	0.0062 ± 0.0002

Table 4. Per-seed FID and KID for all models (full dataset, seeds 42–51).

	FID			**KID**		
Seed	VAE	DCGAN	DDPM	VAE	DCGAN	DDPM
42	140.66	31.10	37.91	0.0315	0.0032	0.0061
43	140.34	29.56	38.35	0.0315	0.0033	0.0065
44	139.59	29.91	37.63	0.0311	0.0031	0.0063
45	140.29	30.29	37.88	0.0315	0.0032	0.0063
46	140.34	30.23	38.05	0.0306	0.0032	0.0063
47	141.28	31.35	37.39	0.0321	0.0033	0.0059
48	140.76	30.87	38.29	0.0315	0.0034	0.0063
49	140.07	30.70	37.20	0.0314	0.0032	0.0060
50	140.26	29.73	38.04	0.0319	0.0030	0.0062
51	140.08	30.82	38.33	0.0311	0.0032	0.0062
Mean	140.37	30.46	37.91	0.0314	0.0032	0.0062
Std	0.43	0.57	0.37	0.0004	0.0001	0.0002

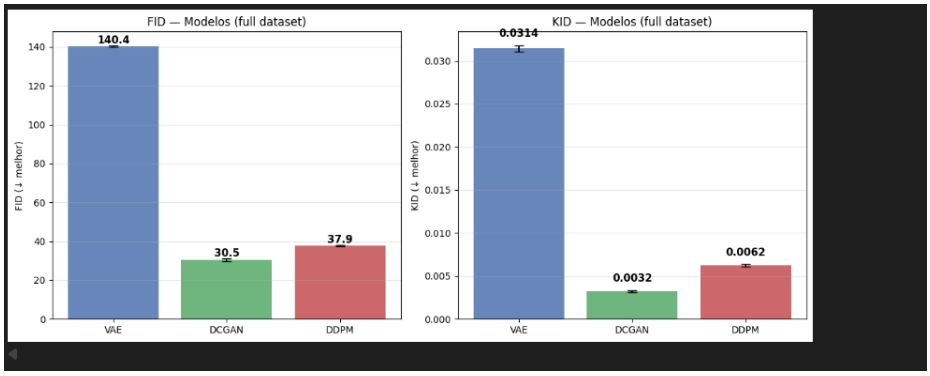

Fig. 8. FID (left) and KID (right) for all models on the full dataset. Error bars represent one standard deviation over 10 seeds.

VAE. The high FID (140.37) reflects the known limitation of pixel-wise BCE: minimising average reconstruction error encourages blurry outputs. The regularised latent space supports smooth interpolation but does not enforce perceptual sharpness.

DCGAN. The adversarial objective directly minimises the perceptual distance between real and generated distributions, yielding the best FID (30.46) and KID (0.0032). At 32×32 resolution the discriminator has sufficient capacity to model the full feature space. The slightly larger FID variance (± 0.57) reflects GAN training stochasticity. Mode dropping is qualitatively visible but not captured by FID/KID.

DDPM. The DDPM ranks second (FID 37.91) with the lowest variance (± 0.37), indicating the most stable generation. The DDIM approximation (100 vs. 1,000 steps) likely accounts for part of the gap with DCGAN. At higher resolutions diffusion models typically surpass GANs [1]; at 32×32 the DCGAN's simpler objective is sufficient.

6.5 Phase 1 vs. Phase 2 — Cross-Phase Comparison

Table 5 quantifies the FID improvement from 20% to 100% training data.

Table 5. FID improvement from Phase 1 (20% subset) to Phase 2 (full dataset). $\Delta\text{FID} = \text{FID}_{100\%} - \text{FID}_{20\%}$.

Model	FID (20%)	FID (100%)	ΔFID
VAE	142.72	140.37	−2.4
DCGAN	47.38	30.46	−16.9
DDPM	45.91	37.91	−8.0

DCGAN benefited most from additional data (ΔFID $= -16.9$), followed by DDPM (-8.0). The VAE showed negligible improvement (-2.4), confirming that its blurriness is a fundamental structural limitation rather than a data-quantity issue.

7 Conclusion

We implemented and compared VAE, DCGAN, and DDPM on the ArtBench-10 dataset under a rigorous two-phase protocol. The DCGAN achieved the best quantitative results on the full dataset (FID 30.46), benefiting most from the larger training set (ΔFID $= -16.9$). The DDPM ranked second (FID 37.91) with the lowest variance and greatest sample diversity, while showing a moderate response to additional data (ΔFID $= -8.0$). The VAE ranked last in both phases, confirming that blur from BCE loss is independent of dataset size.

These results highlight a key trade-off: at 32×32 resolution, adversarial models excel at distribution matching, while diffusion models offer richer diversity and are expected to scale better to higher resolutions [9]. Future work includes full 1,000-step DDPM sampling, class-conditional generation, and evaluation at higher resolutions.

Use of Generative AI Tools. Generative AI Tools like Claude, Gemini and Chat GPT were used as a code assistant during the development of this project, specifically to help structure training loops, debug PyTorch code, and draft parts of this report. All model implementations, experimental results, and conclusions were verified by the authors.

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